

Combating BLE Weak Links by Combining PHY Layer Symbol Extension and Link Layer Coding

Renjie Li , Yeming Li , Jiamei Lv , Hailong Lin , Yi Gao , *Member, IEEE*, and Wei Dong , *Member, IEEE*

Abstract—Bluetooth Low Energy (BLE) technology supports various Internet-of-Things (IoT) applications. However, because of their limited transmission power and channel interference, their performance is deficient over weak links. Extending physical layer symbols or using error correction code to the link layer is effective somehow. Introducing excessive BLE bits to both respectively can also decrease the network throughput. To optimize the BLE technology performance, we propose *CPL*, a combining PHY and link layer optimization technology that adaptively allocates BLE bits to both the physical layer and link layer. Then we propose the *Cross-Layer BLE Bits Dynamic Allocation Model* that unifies the gain of BLE bits in different layers. Finally, we propose an *Interference-Aware Controlled CFO Fine-Tuning Method* that calibrates the model according to different interference patterns. We implement *CPL* on Commercial-Off-The-Shelf (COTS) BLE chips and SDR. The experiment results show that under various interference conditions, *CPL* achieves $50\times$ and 32.16% throughput improvement over RSBLE and Symphony. *CPL* reduces energy consumption by 60.42% to 97.95% compared to RSBLE, and 11.04% to 25.15% compared to Symphony.

Index Terms—IoT, BLE, weak links, cross-layer allocation.

I. INTRODUCTION

BLUETOOTH Low Energy (BLE) supports a variety of Internet-of-Things (IoT) applications including smart homes, wearables, healthcare devices, etc. According to recent research [1], the annual shipment of Bluetooth devices will reach 5.9 billion, including 1.1 billion audio streaming devices, 1.48 billion data transfer devices and 1.03 billion network devices in 2025. However, ubiquitous BLE connectivity often falters in weak links, and due to operating at the crowded 2.4 GHz Industrial Scientific Medical (ISM) frequency [2], [3], other wireless protocols in the 2.4 GHz ISM band (e.g., WiFi at home, ZigBee on factory floors) further degrade its performance. Therefore, an efficient solution for combating BLE weak links is of great importance.

Some existing work uses symbol extension at the physical layer to combat weak links. Symphony [4] generates a single-tone extended symbol by increasing the bits in the BLE packet,

which we define as BLE bits ($160\times$). ChirpTransformer [5] adaptively selects different extension schemes based on different noise levels. Some other work uses coding methods at the link layer to combat weak links. RSBLE [6] integrates the native BLE with Reed-Solomon (RS) coding, and HOPE [7] uses rateless coding schemes to enhance throughput by adapting to channel qualities of potential receivers. Hope can dynamically increase the BLE bits for link layer coding to improve the reliability of data transmission under time-varying channel conditions.

Although much work has been done, the question remains: when is it more beneficial to use the physical layer and when is it more advantageous to use the link layer? To answer this question, we conduct a preliminary experiment, and some interesting findings emerged: For the physical layer, as we gradually increase the symbol extension multiple, we find that the returns from adding extra BLE bits for extension exhibit the law of diminishing returns. Specifically, when 1-bit BLE symbols are extended to 2 bits, the decrease in Bit Error Rate (BER) is 15.51% . However, when 9-bit BLE symbols are extended to 10 bits, the decrease in BER is only 0.46% . This indicates that the benefits of adding extra BLE bits for extension decline with the increase of the extension multiple. For the link layer, under very low Signal-to-Noise Ratios (SNR), using BLE bits for error correction proves to be inefficient. The reason is that it becomes difficult to identify the unextended preamble and access address of BLE packets during packet detection, leading to packet discard. Compared to the symbol extension technique, this can result in more than 50% throughput loss, which severely limits the coding gain and wastes the extra BLE bits used for error correction. Therefore, under different conditions, the same length of BLE bits added to the physical layer does not result in the same gain compared to adding them to the link layer. In order to allocate BLE bits more efficiently to combat BLE weak links, we propose *CPL*, a low power wireless protocol performance optimization method Combining PHY and Link layer. Considering changing channel conditions, including time-varying noise intensity and various interference patterns, efficient and fast allocation of BLE bits for physical layer symbol extension and link layer coding remains a challenge.

First, the time-varying noise intensity leads to a time-varying SNR. Existing methods, such as 4C [8], can calculate the throughput of a network with different allocation schemes under current channel conditions through data collection, offline training, and online prediction. However, this machine learning approach requires a large amount of data, while data collection and processing are difficult in the IoT field. Because of the

Received 12 April 2025; accepted 11 June 2025. Date of publication 16 June 2025; date of current version 3 September 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62272407, in part by the “Pioneer” and “Leading Goose” R&D Program of Zhejiang under Grant 2023C01033, in part by the National Youth Talent Support Program, and in part by Ningbo Yongjiang Talent Project. Recommended for acceptance by F. Liu. (Corresponding authors: Jiamei Lv; Wei Dong.)

The authors are with the College of Computer Science, Zhejiang University, Hangzhou 310027, China (e-mail: lirj@zju.edu.cn; liymemnets@zju.edu.cn; lvjm@zju.edu.cn; linhl@zju.edu.cn; gaoy@zju.edu.cn; dongw@zju.edu.cn).

Digital Object Identifier 10.1109/TMC.2025.3579934

large number of BLE devices and their wide distribution, data collection costs are high, and data quality is spotty. The approach for addressing the data collection challenge is TALENT [9], which employs an online learning algorithm to dynamically train its model without the prerequisite of collecting a large and comprehensive training dataset, unlike the 4C method. TALENT's online learning algorithm, while beneficial for adaptability in IoT networks, suffers from the significant time overhead of continuous and periodic weight updates, leading to increased power consumption, reduced battery life, and potential latency issues in time-sensitive applications. Another method is DPLC [10], which dynamically adjusts the packet length based on a link estimation method that passively observes packet reception. However, this method requires a long convergence time and cannot adapt to the fast-changing channel conditions in a timely manner.

Therefore, we establish a *Cross-Layer BLE Bits Dynamic Allocation Model* to select an efficient scheme adapted to the time-varying channel quickly by offline modeling and online allocation. The offline modeling comprehensively considers the physical and link layer characteristics and calculates the throughput of various allocation schemes at any SNR as a basis for online selection of the allocation scheme. There are two primary challenges: first, model construction is inherently difficult due to the fact that the expressions differ under various SNRs and are not easily represented linearly, making pure white-box modeling approaches impractical. Second, employing a purely black-box approach incurs significant overhead, as the expressions vary under different network conditions, and there is a lack of sufficient datasets for comprehensive collection. To address these challenges, we propose a hybrid methodology that combines white-box modeling with a lightweight black-box approach, thereby facilitating a more feasible and efficient solution. Since our method does not require a large amount of data collection, periodic weight update and long convergence time, it can efficiently and fastly allocate BLE bits on resource-constrained devices to optimize network throughput and energy consumption.

Second, a BLE device encounters a wide variety of noise patterns during its operation. Considering that the noise of different patterns in weak links affects the accuracy of the model, a fixed model is difficult to adapt to various noise patterns. The straightforward approach is to install all the optimizers (i.e., models) corresponding to different noise patterns in a single device. However, in resource-constrained devices (e.g., 1 MB of flash in nRF52840 [11] and 256 kB of flash in cc2540 [12]), multiple optimizers can result in wasted storage space, even beyond their remaining storage; and the maintenance and upgradation of all the optimizers pose a continuous cost and technical challenge as technology evolves and noise patterns change. Therefore, we propose an *Interference-Aware Controlled CFO Fine-Tuning Method* to adapt our model to various noise patterns. The method analyzes the channel spectrum in real-time to infer the noise patterns and utilizes the controlled carrier frequency offset (CFO) of the BLE devices to actively mitigate the narrow-band interference.

We implement *CPL* with Commercial-Off-The-Shelf (COTS) BLE chips and Universal Software Radio Peripheral (USRP). The experiment results show that in the presence of broadband interference (e.g., WiFi signal) when comparing *CPL* to RSBLE, the range of throughput enhancement is $1.25\times$ – $50.56\times$. The throughput improvement of *CPL* relative to Symphony ranges from 5.49% to 18.78%. In the presence of narrowband interference (e.g., ZigBee signal), the throughput improvement of *CPL* over RSBLE ranges from $5.06\times$ to $27.25\times$. When compared to Symphony, the improvement of *CPL* ranges from 25.54% to 32.16%. *CPL* reduces energy consumption by 60.42% to 97.95% compared to RSBLE, and 11.04% to 25.15% compared to Symphony for transmitting 255B data.

The contribution of this paper is three-fold:

- We propose *CLP* and construct an efficient and fast optimization model, *Cross-Layer BLE Bits Dynamic Allocation Model*, to enhance network throughput over weak links by adaptively allocating BLE bits at physical and link layers. With *CLP*, existing COTS BLE chips can quickly and reliably transmit the data to the receiver.
- In order to adapt the *CPL* to different noise patterns, we propose an *Interference-Aware Controlled CFO Fine-Tuning Method*. The method dynamically evaluates the channel spectrum, infers the noise patterns, and utilizes the controlled CFOs of the BLE devices to proactively mitigate the narrowband interference. The model is fine-tuned to ensure that an efficient allocation of BLE bits is achieved to improve the performance of weak links.
- We implement the *CPL* system with COTS BLE chips and Software-Defined Radio (SDR) technology. Extensive experiments are conducted under various noise intensities and types of interference. The results demonstrate that *CPL* can significantly enhance the performance (e.g., throughput and energy consumption) of BLE transmissions compared to existing works. Furthermore, through a specific experimental case study, we illustrate how *CPL* adapts to the current environment with reasonable time to achieve an best in changing channel conditions.

The remaining of the paper is organized as follows: Section II introduces the related work and motivation of the paper. Section III introduces the overview of our system *CPL*. Sections IV and V further respectively detail *Cross-Layer BLE Bits Dynamic Allocation Model* and *Interference-Aware Controlled CFO Fine-Tuning Method*, as two main parts of the system. Section IV-B introduces the implementation of the *CLP* system along with the experimental setup. Section VII describes the experiments and discusses the results. Finally, Section VIII concludes this work.

II. RELATED WORK AND MOTIVATION

Physical Layer Symbol Extension: Short symbols of low-power wireless protocols can be easily overwhelmed by noise and interference. Symbol extension can increase the energy of a single symbol, thereby enabling correct symbol demodulation at low SNR levels. Z. Li et al. propose a symbol extension

technology [4] for BLE and ZigBee to extend the communication range. By generating specific patterns of signals (i.e., single-tone sine waves) on BLE and ZigBee devices, LoRaWAN base stations can detect these signals using their highly sensitive receivers, enabling long-distance communication. Y. Cao et al. propose Mrs. Z [13], a novel physical layer design that achieves multi-rate selection in ZigBee by varying the single spectrum spreading length to multiple lengths. This design predicts the potential impact of bit errors on throughput based on different error types and selects the transmission rate accordingly. C. Li et al. design the ChirpTransformer framework [5], which enhances network coverage, throughput, and energy efficiency for LoRa nodes by introducing four different chirp features. These features include the airborne time of symbols, selectable initial frequencies, chirp repetition, and symbol transitions, collectively providing a wider range of encoding options for LoRa nodes. While these studies explore numerous schemes for optimizing communication performance at the physical layer, they focus solely on enhancements within the physical layer and do not integrate optimizations with other hierarchical structures of the OSI model.

Link layer coding: Link layer coding is the key technology to ensure reliable data transmission in wireless networks. Şişman. Cem et al. conduct an in-depth analysis of the performance of BLE and ZigBee in terms of coding gain [6]. Convolutional coding, RS coding and BCH coding can improve the reliability of low-power radio data transmission to some extent. However, the coding acts only on the payload and does not involve error correction of the preamble and access address, which can lead to packet detection failure and packet loss at low SNR. Excessively high packet loss rate can significantly limit the gain of coding. Additionally, Z. Li et al. introduce a hopless network paradigm [7] utilizing the latest rateless coding schemes to enhance throughput by adapting to channel qualities of potential receivers regardless of their hop distances. Likewise, these studies are confined to considerations within a single-layer structure, limiting the scope for performance optimization.

Motivation: Both physical layer expansion and link layer coding techniques have their respective benefits in enhancing the network throughput of weak link BLE. To determine when to use physical layer expansion and link layer coding techniques for maximum benefit, we further explore the characteristics and limitations of these two methods. In terms of the physical layer, we gradually increase the symbol extension multiple to investigate a decrease in BER with each additional bit, as shown in Fig. 1. When 1-bit BLE symbols are extended to 2 bits, the decrease in BER is 15.51%, while the decrease in BER of 5-bit to 6-bit and 9-bit to 10-bit are only 1.51% and 0.46%, far below that of 1-bit to 2-bit. We find that the benefits of adding extra bits in a BLE packet (defined as BLE bits) for symbol extension decrease with the increase of the extension multiple, exhibiting a diminishing returns effect. For link layer coding techniques, we discover that although using a certain proportion of BLE bits as error correction codes can efficiently correct errors, under extremely low signal-to-noise ratios, the preamble and access address of BLE packets are not extended, making it difficult to identify the preamble and access address during

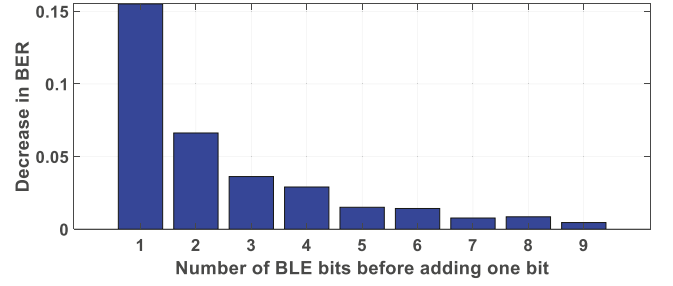


Fig. 1. Decrease in BER with each additional bit for symbol extension.

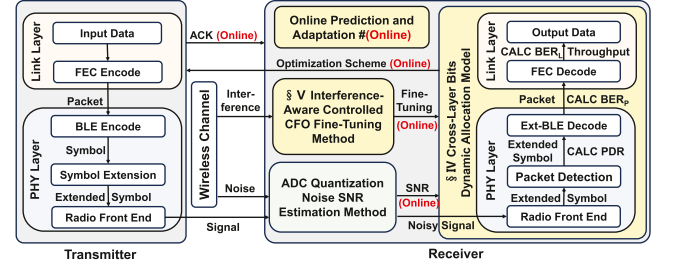


Fig. 2. Overview of system.

packet detection, resulting in packet discard. In such cases, compared to the physical layer symbol extension technique, this can cause more than 50% performance loss, which greatly limits the coding gain of the link layer and wastes the additional BLE bits used for error correction.

Since both methods have their strengths and weaknesses and can achieve the highest benefits under different circumstances, it motivates the design of a bit-efficient cross-layer method to combat BLE weak links.

III. SYSTEM OVERVIEW

CPL is designed with a focus on the cross-layer interaction between the physical layer and the link layer of BLE technology to enhance its performance in weak links. This section introduces the design and implementation of our **CLP** system, elucidating the pivotal system modules and practical implementation of our research. Fig. 2 shows the overview of our system, which contains the following key components:

Cross-Layer BLE Bits Dynamic Allocation Model: A model that unifies the gain of adding BLE bits in different layers, allowing adaptive allocation of resources to enhance performance over weak links. This model calculates throughput under various BLE Bits allocating schemes across the PHY layer and link layer with a high degree of accuracy between modeling and actual performance. This model also supports the system's ability to optimize network throughput because it calculates a good allocation scheme that adapts to the current link conditions to efficiently allocate resources on weak links and improve network throughput.

Interference-Aware Controlled CFO Fine-Tuning Method: A method for assessing channel conditions and calibrating the system model to find the efficient BLE bits allocation, which

maximizes throughput under varying interference patterns. The CFO fine-tuning method of interference-aware control ensures the robustness of the system under interference. It intelligently evaluates and responds to changing channel conditions for reliable communication. The CPL system also focuses on the needs of resource-constrained equipment. Its offline training and online prediction strategies allow these devices to quickly adjust BLE bits allocation based on current link conditions, ensuring efficient operation with minimal computational effort.

Online Prediction and Adaptation: An online Feedback mechanism that enables dynamic adaptation to evolving channel conditions, ensuring efficient performance over weak links with minimal computational overhead. This mechanism relies on a collaborative process between the transmitter and the receiver to exchange information about the channel state and the applied allocation scheme. The receiver plays a crucial role by periodically monitoring the channel quality metrics (SNR and interference patterns). When the measured change in channel conditions surpasses a predefined threshold, indicating a need for a new allocation scheme, the receiver encapsulates the updated scheme information within a BLE data packet and transmits it to the transmitter. The receiver then awaits an Acknowledge character (ACK) packet from the transmitter to confirm successful reception of the updated scheme information. Upon receiving the BLE packet, the transmitter switches its Ubertooth from *MODE_TX_SYMBOLS* to *MODE_RX_SYMBOLS* to receive the packet, decodes the BLE packet, and updates its allocation scheme accordingly. Subsequently, the transmitter switches its Ubertooth back to *MODE_TX_SYMBOLS* to prepare for data transmission using the new allocation scheme. Before initiating data transmission with the new scheme, the transmitter sends a BLE packet using the old scheme to inform the receiver of the switch, allowing the receiver to utilize the updated scheme for decoding the incoming signal.

IV. CROSS-LAYER BLE BITS DYNAMIC ALLOCATION MODEL

This section aims to comprehensively consider the characteristics of the physical and link layers, and incorporate the concept of offline modeling as well, to enhance the overall system performance (i.e., network throughput) in the context of exploring cross-layer optimization strategies for BLE networks. Through offline modeling, we can analyze and simulate various scenarios based on the features of the physical and link layers in advance, which may provide valuable insights and references for formulating more effective cross-layer optimization strategies to ultimately boost the network throughput of BLE networks. Throughput is influenced by various factors at both the physical and link layers, necessitating a comprehensive optimization approach. Subsequent subsections will sequentially introduce the BER and Packet Detection Rate (PDR) modeling of the BLE physical layer, and the structure and error correction capability of the RS coding scheme at the link layer. The specific parameters of the Modeling of the physical and link layer are shown in Table I. Finally, we will discuss how to integrate the models of the physical and link layers to identify cross-layer BLE bits allocation schemes that achieve maximum throughput

TABLE I
MODELING PARAMETERS OF PHYSICAL AND LINK LAYER

Symbol	Parameters
γ	Signal-to-noise ratio in the case of GFSK modulation.
BER	Bit error rate in the case of GFSK modulation.
h	Modulation index of GFSK.
$Q_1(\cdot, \cdot)$	First-order Marcum Q-function.
$I_0(\cdot)$	0-order modified Bessel function.
ρ	Sampling function concerning $2\pi h$.
$g(t)$	Impulse response of the Gaussian filter.
Δ	Impact of the extension factor on the BER.
P_{As}	Probability of successful preamble and AA acquisition.
PDR	Packet detection rate for BLE communication.
$d_{\min}$	Code minimum distance.
R_r	Symbol error rate after RS decoding.

across different SNR ranges. This integration will demonstrate how cross-layer optimization strategies can enhance the overall performance of BLE networks.

A. Physical Layer Modeling

1) BER Modeling: This part models the BER within the physical layer of BLE communication. Initially, by analyzing the characteristics of Gaussian Frequency Shift Keying (GFSK) modulation, we model the BER of the traditional BLE physical layer. Then, we introduce the general setup for BLE communication. Finally, we conduct a fitting process to obtain a novel correction formula for the impact of the BLE symbol extension.

Let $\text{BER}(\gamma)$ denote the BER obtained for the transmission rate $R = 1$ Mbps, with a SNR $\gamma = E_s/N_0$. According to [14], the BER in the case of GFSK modulation is given by

$$\text{BER}(\gamma) = Q_1(a, b) - \frac{1}{2} \exp\left(-\frac{a^2 + b^2}{2}\right) I_0(ab), \quad (1)$$

where $Q_1(\cdot, \cdot)$ is the first-order Marcum Q-function, and $I_0(\cdot)$ is the 0-order modified Bessel function. a and b can be expressed as:

$$a = \sqrt{\frac{\gamma}{2} (1 - \sqrt{1 - \rho^2})}, b = \sqrt{\frac{\gamma}{2} (1 + \sqrt{1 - \rho^2})}. \quad (2)$$

ρ is a sampling function concerning the modulation index h of GFSK,

$$\rho = \sin(2\pi h)/2\pi h. \quad (3)$$

So $\text{BER}(\gamma)$ can be further rewritten as

$$\text{BER}(\gamma) = \exp\left(\frac{-\gamma}{2}\right) \left\{ \frac{1}{2} I_0(ab) + \sum_{k=1}^{\infty} \left(\frac{a}{b}\right)^k I_k(ab) \right\}, \quad (4)$$

where $I_k(\cdot)$ is the k -order modified Bessel function.

In BLE communications, each data packet is assumed to traverse an independent, static Additive White Gaussian Noise (AWGN) channel. The simulation setup encompasses components of a transmitter, wireless channel, and receiver. Upon completion of packet assembly, the binary data is mapped onto a Non-Return-to-Zero (NRZ) coding scheme at a rate of 1 Mbps.

Subsequently, these NRZ codes are processed through a Gaussian filter, which involves convolution with its impulse response. The BER operates with a bandwidth of 2 MHz. The gross bit rate achieves 1 Mbps within a 2 MHz channel bandwidth through GFSK modulation with $BT = 0.5$ and a modulation index ranging between 0.495 and 0.505. The impulse response of the Gaussian filter is specified as follows:

$$g(t) = \text{Exp}(-t^2/2s^2T^2) / \sqrt{2\pi}sT. \quad (5)$$

Considering BLE physical layer extension, which refers to the prolongation of the BLE symbol's duration, the extension factor is denoted by ext , representing the multiple by which the symbol is extended. The extension of physical layer symbols leads to a multiplication of the total phase difference, thereby significantly enhancing the noise immunity performance of the physical layer signal.

Due to the absence of prior work involving the modeling of the BLE physical layer symbol extension, this paper pioneers an attempt to conduct a modeling analysis. A straightforward approach is to fit the BER versus SNR curves for each extension factor individually. However, employing polynomial fitting results in excessive computational complexity, while simple linear fitting lacks sufficient precision. Preliminary experiments, as illustrated in Fig. 1, reveal that the benefits of symbol extension diminish with an increase in the extension multiple, manifesting a clear diminishing returns effect. This observation provides additional information for a more efficient and accurate modeling approach. We define the benefits of symbol extension as G_{add} , which quantifies the incremental improvement in SNR attributable to the symbol extension process. This metric captures the net enhancement in signal quality, facilitating a more precise characterization of the relationship between symbol extension and communication performance. Through the analysis of preliminary experiment results, the additive SNR gain G_{add} brought about by symbol extension can be represented as a logarithmic function of ext :

$$G_{add} = c \ln(d \cdot ext), \quad (6)$$

where c and d are coefficients to be determined, which can be obtained through fitting experimental data. The diminishing marginal returns imply that the derivative of G_{add} concerning ext gradually decreases. We compute the derivative of G_{add} :

$$\frac{dG_{add}}{dext} = \frac{c \cdot d}{d \cdot ext} = \frac{c}{ext}, \quad (7)$$

where the derivative decreases as ext increases, aligning with the observation of diminishing marginal returns. Further refinement of the model is achieved through extensive fitting of experimental data, enabling the precise determination of the coefficients c and d . This iterative process reveals a more nuanced relationship between ext and G_{add} , leading to the refined expression:

$$G_{add} = \frac{4ext + 8}{ext} \ln ext = \frac{ext + 2}{ext} \ln ext^4. \quad (8)$$

The update for γ is changed to γ_e ,

$$\gamma_e = \gamma + G_{add} = \gamma + \frac{ext + 2}{ext} \ln ext^4. \quad (9)$$

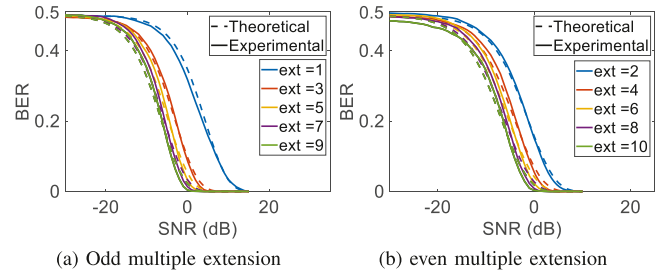


Fig. 3. BER-SNR relationships with symbol extension.

The update for a and b is changed to a_e and b_e ,

$$\begin{aligned} a_e &= \sqrt{\frac{\gamma_e}{2} (1 - \sqrt{1 - \rho^2})}, \\ b_e &= \sqrt{\frac{\gamma_e}{2} (1 + \sqrt{1 - \rho^2})}. \end{aligned} \quad (10)$$

So BER_e can be further rewritten as

$$\begin{aligned} \text{BER}_e &= \exp\left(\frac{-\gamma_e}{2}\right) \left\{ \frac{1}{2} I_0(a_e b_e) \right. \\ &\quad \left. + \sum_{k=1}^{\infty} \left(\frac{a_e}{b_e}\right)^k I_k(a_e b_e) \right\}. \end{aligned} \quad (11)$$

To facilitate a clearer comparison between the simulation experiments and the formulaic modeling for alignment, we segregate the extension factor into odd and even values to juxtapose the outcomes, as illustrated in Fig. 3. The results indicate that in all the extension factor scenarios considered, the empirical findings closely align with the predictions of formulaic modeling. Consequently, it is feasible to employ the formulaic modeling approach within an acceptable degree of inaccuracy. Furthermore, it is observed that although the noise immunity of the physical layer intensifies with an increase in the extension factor, this enhancement is not linear. When the extension factor ext increases from 1 to 3, the improvement in noise immunity for BLE is significantly greater than the gain achieved when ext increases from 7 to 9. Therefore, when allocating redundant resources to the physical layer, it is essential to consider whether the benefits of adding BLE bits are justified.

2) *PDR Modeling*: This part introduces the modeling of the PDR within the physical layer of BLE communications, detailing the process of signal synchronization, preamble, and Access Address (AA) acquisition, and the impact of symbol extension on the robustness of packet detection in noisy conditions.

AA is a 32-bit pseudo-random sequence that serves as a synchronization code for packet detection and is crucial for establishing communication between devices. The preamble, which precedes the AA, is an 8-bit known sequence of bits that aids in initial signal synchronization. The acquisition of the AA and preamble involves the following steps.

- 1) *Low-Pass Filtering*: Apply a low-pass filter to the raw signal to reduce high-frequency noise.
- 2) *Phase Difference Calculation*: Compute the phase difference between consecutive samples of the filtered signal.

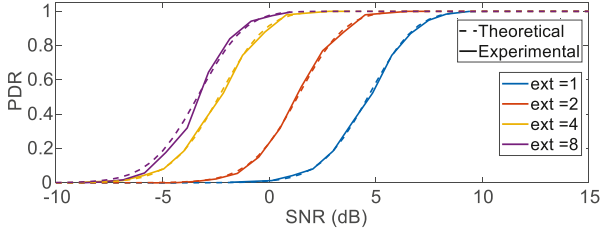


Fig. 4. PDR-SNR relationships with symbol extension.

- 3) *Correlation Operation*: Perform a correlation using a pre-defined mask that indicates the expected phase differences for the known preamble and AA sequence.
- 4) *Peak Detection in Correlation Output*: Analyze the correlation results to identify peaks, which signify potential synchronization points within the signal.
- 5) *Synchronization Point Identification*: Determine the starting position of the preamble and AA by these peaks.
- 6) *Signal Synchronization*: Upon detecting the preamble and AA, the receiver synchronizes with the signal.

We define P_{A_s} as the probability of successful acquisition for preamble and AA, which can be expressed as

$$P_{A_s} = \sum_{k=0}^S \binom{32}{k} BER_e^k (1 - BER_e)^{32-k}, \quad (12)$$

where BER_e is given in (11).

Based on the identified start positions of the preamble and AA, the signal is partitioned into discrete segments. The segmented signals are subjected to a Fast Fourier Transform (FFT) to decode the results. This transformation translates the time-domain signal into the frequency domain, enabling the extraction of the embedded data. Following the decoding of the segmented signals, a verification process is undertaken to ascertain the fidelity of the decoded data. This involves cross-referencing the decoded output against the expected AA sequence. If the decoded data and the anticipated AA sequence are well-matched, the packet will be correctly detected. We define PDR_e as the packet detection rate under the impact of BLE symbol extension, expressed as:

$$PDR_e = P_{A_s} \left[(1 - BER_e)^3 + 3BER_e (1 - BER_e)^2 \right]^{18}. \quad (13)$$

It is worth noting that the BLE symbol extension at the physical layer can effectively enhance the PDR of data packets. Fig. 4 illustrates how the symbol extension at the physical layer improves the PDR in noisy environments. During the packet detection process, the accurate identification of the preamble and access address is crucial. If they are not correctly identified, the packet will be discarded, leading to a decrease in the overall throughput. Therefore, the symbol extension of the packet's preamble and access address can improve the P_{A_s} and reduce the BER in the weak link, thereby enhancing the robustness of the packet detection process.

B. Link Layer Modeling

This subsection delves into the modeling of the link layer coding scheme (e.g., RS codes), exploring their structural attributes and error-correction capabilities.

In the field of error correction, RS codes [15] are a widely utilized class of codes at the link layer. Due to their remarkable robustness and versatility, RS codes are widely used in a variety of applications. RS codes are distinguished as nonbinary cyclic codes, employing symbols composed of m -bit sequences. The structure of RS (n, k) codes on m -bit symbols is crafted to be universally applicable, catering to all possible combinations of n and k .

$$k < n < 2^m + 2, \quad (14)$$

where k is the number of data symbols being encoded. n is the total number of code symbols in the encoded block. m signifies any integer greater than 2, marking the length of the bit sequence. n and k define the code's construction and error-correction capacity.

$$(n, k) = (\eta, \eta - 2c), \quad (15)$$

where c is the symbol-error-correcting capability of the code. $n - k = 2c$ is the number of parity symbols. η denotes $2^m - 1$.

RS codes can attain the maximum achievable code minimum distance within the realm of linear codes that possess identical input and output block lengths. In nonbinary codes, the term 'distance' is utilized to denote the divergence between two code-words, measured by the count of symbols where the sequences exhibit discrepancies. For RS codes, the code minimum distance is given by [16]:

$$d_{\min} = n - k + 1. \quad (16)$$

RS codes excel at correcting burst errors. Their efficiency is particularly evident in scenarios where the channel supports a large alphabet of input symbols. A notable characteristic of RS codes is their capacity to incorporate up to two additional information symbols into an existing code of length n , without compromising its minimum distance. This extension results in a new code of length $n + 2$ while retaining the original count of parity symbols. The Symbol Error Rate (SER) after RS decoding, denoted as R_r , can be expressed in relation to the SER of channel R_c [17].

$$R_r \approx \frac{1}{\eta} \sum_{j=c+1}^{\eta} j \binom{\eta}{j} R_c^j (1 - R_c)^{\eta-j}. \quad (17)$$

The symbols are made up of m bits each. According to [17], the BER can be upper bounded by the SER for specific modulation types. When comparing the theoretical and exponential relationship of the BER of the link layer after RS decoding P_B and the BER of the physical layer, a visualization is presented in Fig. 5. The difference between the actual and theoretical values is within an acceptable range.

C. Cross-Layer Optimization

This subsection delineates our approach to cross-layer optimization in BLE networks. To enhance the overall performance

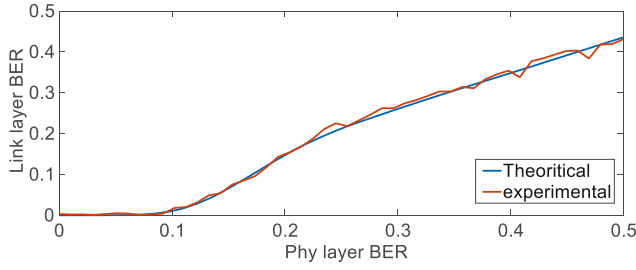


Fig. 5. Relationship of the link layer BER and physical layer BER.

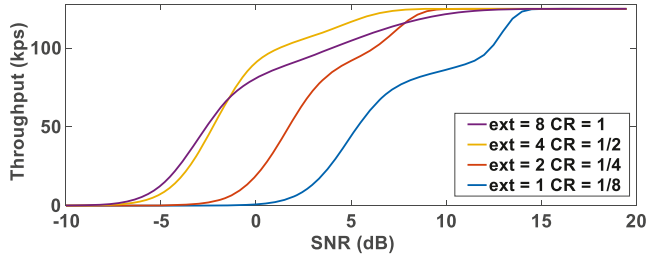


Fig. 6. Throughput across different allocation schemes (an example of $8 \times$ BLE bits for physical layer and link layer).

of BLE systems, we implement a cross-layer optimization strategy that integrates the physical and link layers. The primary objective is to maximize network throughput, which is a critical metric for evaluating the efficiency of data transmission in wireless networks. Network throughput Th , influenced by various factors across both the physical and link layers, can be calculated by the (18).

$$Th = \frac{N * (1 - BER_L) * PDR_e * CR}{T * ext}, \quad (18)$$

where N is the total amount of transmitted data. T signifies the total duration of data transmission. BER_L is the BER output by the link layer decoding (after PHY layer symbol extension). PDR_e is the packet detection rate after optimization of the PHY layer symbol extension. CR is the code rate, a fractional number that expresses what part of the redundant message is actually meaningful. For example, if the code rate is $4/7$, for every 4 bits of useful information, the coder generates a total of 7 bits of data, of which 3 bits are added from BLE bits. ext is the physical layer expansion factor. When we allocate more BLE bits at the physical layer, ext will be larger.

To achieve cross-layer optimization, it is necessary to integrate the models of the physical layer and the link layer. The model of the physical layer focuses on the BER of demodulation and data packet detection. CR and ext are both pre-set according to the setup of different schemes during the experimental process. Regarding N and T , their ratio represents the data transmission rate of 1Mbps (LE 1M PHY) [18]. The calculation of throughput in Fig. 6 is theoretical results from (18) as well as the modeling results of the physical layer and link layer (Section IV-A and IV-B). Additionally, it is worth noting that the throughput of the experiments in Section VII are calculated

based on actual measurements of BER_L and PDR along with the preset CR and ext , according to (18).

We leverage BLE symbol extension techniques to reduce the BER and enhance the PDR at the physical layer. On the other hand, the model of the link layer is concerned with error correction. In this regard, we mainly employ the RS code as an efficient forward error correction (FEC) code. Combining (11), (13), and (17) from Section II, we derive the curve illustrating the variation of the network throughput with respect to different variables. Fig. 6 shows an example of eight times the total number of BLE bits for the physical layer and link layer.

We observe that when the SNR rises from a low to high level, the concavity of the throughput curve in Fig. 6 changes several times. The change in the SNR leads to shifts in the dominant factors that constrain the throughput, thereby influencing its concavity characteristics. For instance, in environments with low noise levels, packets are detected effectively with low packet loss rates. In such scenarios, the BER of the packet payload becomes the primary limiting factor for throughput. Conversely, in environments with high noise levels, difficulties in accurately identifying the preamble and access address within packets lead to a significant decrease in PDR. Therefore, the principal factor that limits the throughput changes to the PDR. Within various SNR ranges, the best cross-layer BLE bits allocation schemes corresponding to the maximum throughput also differ.

In our study, the SNR of the BLE nodes is meticulously determined by employing the Analog to Digital Converter (ADC) Quantization Noise SNR Estimation Method. This method is inspired by the work [19]. Utilizing the FFT in conjunction with a Kaiser window function, we estimate the power spectral density of the signal and compute the average power of the main frequency band and its harmonics. Concurrently, we estimate the noise power by zeroing the frequency bands outside the main band and calculating the median of the remaining spectrum. The processing gain of the FFT, as a frequency-domain analysis tool, plays a critical role in SNR analysis. Specifically, the amplitude of the signal increases by the square root of N_{FFT} , while the power increases by N_{FFT} squared. Here, N_{FFT} denotes the number of FFT samples. In contrast, the average power of the noise increases only by N_{FFT} . Consequently, the theoretical gain of FFT is given by $10\lg(N_{FFT})$. However, the increase in signal power and spectral leakage caused by the FFT may lead to an overestimation of the computed SNR. To mitigate this effect, we subtract $10\lg(N_{FFT})$ when correcting the signal power, and we divide the noise power of each FFT bin by N_{FFT} to obtain an accurate noise power estimate when calculating the output SNR. In practice, we perform an FFT transformation on the signal, compute and correct the spectral power of both the signal and the noise, and ultimately derive an accurate SNR value unaffected by the FFT processing gain, thereby providing a more genuine assessment of the noise robustness and performance optimization of signal processing algorithms.

Leveraging the relationship curve in Fig. 6 along with other multiple such as $4 \times$, $16 \times$, and $32 \times$, we can identify cross-layer BLE bits allocation schemes that achieve best throughput over a various range of SNRs.

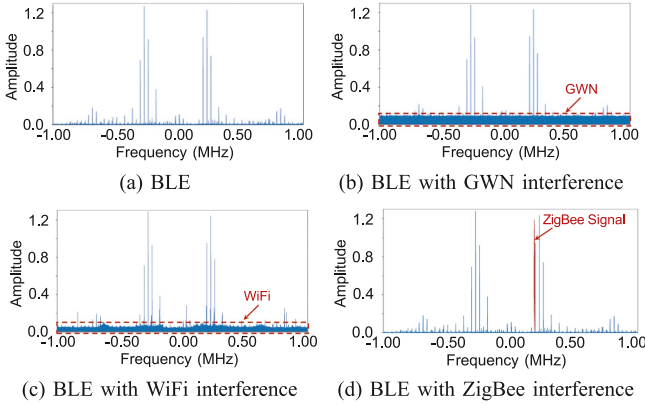


Fig. 7. The spectrum graphs of BLE and interference.

V. INTERFERENCE-AWARE CONTROLLED CFO FINE-TUNING METHOD

In the dynamic landscape of real-world wireless communications, BLE devices are invariably subjected to a myriad of interference patterns that can significantly impair their performance. Apart from the ambient noise environment that can be approximated as Gaussian White Noise (GWN), there are also common interference such as broadband and narrowband signals that can disrupt our wireless devices. This section delves into the nuanced impact of these interferences on BLE networks, with a particular focus on how they affect the spectral characteristics of wireless signals. Furthermore, we propose an innovative Interference-Aware Controlled CFO Fine-Tuning Method, which not only distinguishes between different types of interference through frequency-domain analysis but also leverages the controllable Carrier Frequency Offset (CFO) of BLE devices to enhance the robustness of BLE communications against diverse interference scenarios.

Since we primarily employ spectrum analysis to understand the wireless channel environment and calculate the current SNR by *ADC Quantization Noise SNR Estimation Method*, it is imperative to investigate how interference specifically impacts the spectral characteristics of wireless signals. Next, we mainly analyze the impact of broadband interference and narrowband interference (taking WiFi and ZigBee signals as examples). Fig. 7 illustrates the spectrograms of BLE signals and their corresponding characteristics in various interference patterns.

Analyzing Fig. 7(c), we observe that the bandwidth of WiFi interference is significantly larger than that of the BLE signal. Therefore, within the frequency band considered for BLE spectrum analysis, WiFi interference can be approximated as the influence of GWN shown in Fig. 7(d). Analyzing Fig. 7(d), we notice that narrowband interference has a more pronounced impact on BLE spectrum analysis, falling within the bandwidth range of BLE. These observations highlight the diversity in interference patterns, such as the broadband nature of WiFi and the narrowband characteristics typical of other low-power wireless protocols, which can affect the BLE signal reception and analysis.

Therefore, we propose a *Interference-Aware Controlled CFO Fine-Tuning Method* utilizing the controllable CFO of BLE devices to mitigate narrowband interference with proximity to the boundary actively. First, we adopt a frequency-domain analysis technique used to differentiate between types of interference. Specifically, we eliminate the peaks corresponding to BLE signal frequencies (known in advance) in the spectrum obtained after applying FFT to the signal, thereby retaining the patterns of the interference signals. We then compare the bandwidth of the interference signals with the defined bandwidth ranges for narrowband and wideband interference signals to determine whether it is wideband or narrowband interference. Here, the defined ranges can be adjusted according to different channel environments, and in this paper, signals with a bandwidth of less than 6 MHz are considered narrowband signals.

CFO is the difference in frequency between the carrier frequencies of the transmitter and the receiver. Prior to transmission, the baseband interference is modulated onto the carrier wave with frequency f_{ctx} , and the BLE signal in the air is represented as $e^{j[\phi(t)+2\pi f_{\text{ctx}}t]}$. Upon reception, the receiver eliminates the carrier wave using its local oscillator with frequency f_{ctx} , resulting in the received baseband signal being $e^{j[\phi(t)+2\pi f_{\text{CFO}}t]}$, where $f_{\text{CFO}} = f_{\text{ctx}} - f_{\text{ctx}}$. Due to hardware imperfections, the carrier frequencies of individual devices are different. The center frequency of the received baseband signal contains a frequency offset rather than being zero, which is referred to as the CFO. CFO is commonly present in 2.4 GHz low-power wireless technologies. BLE demonstrates a broad tolerance for CFO, up to 150 kHz or ± 62.5 ppm. In contrast, the IEEE 802.15.4 standard defines a maximum CFO of ± 40 ppm. The fractional CFO can reach up to ± 42 kHz according to the IEEE 802.11n standard [20], [21], [22]. Given the varying upper limits of CFO specified by different low-power wireless protocols, it is essential to properly configure CFO-related parameters to better implement the interference-aware controlled CFO fine-tuning method.

VI. IMPLEMENTATION

In the subsequent paragraphs, we introduce the practical implementation of our research process, detailing the hardware and software configurations. We elaborate the selection and setup of the transmitter, receiver units, and interference sources, as well as the specific experimental site chosen for our tests.

Transmitter: We select the Ubertooth One [23] as our off-the-shelf hardware solution for Bluetooth Low Energy applications. This device incorporates an NXP LPC1756 [24] MCU along with a TI CC2400 [25] chip designed for BLE communication. The CC2400 is capable of delivering an output power at 0 dBm, and when integrated with the external power amplifier [26] featured on the Ubertooth One, it achieves a 20 dB gain. Consequently, the resultant output power level is approximately 20 dBm [27], which is in full compliance with the Bluetooth standard's output power range of -20 to 20 dBm [28]. We fix the transmission power of BLE at 0 dBm and only alter the cross-layer BLE bits allocation scheme employed during the

transmission process, specifically at the physical layer and link layer.

Receiver: The implementation of the *CPL* receiver functionality is achieved by utilizing the USRP B210, a high-performance software-defined radio platform. This platform is interfaced with a portable computational device equipped with an Intel Core i7-8565 U processor and NVIDIA GTX 3090 graphics card, which serves as the data processing task. The system operates on a channel centered at a frequency of 2.39 GHz to detect advertising packets emitted by BLE devices configured to the identical frequency. Such a specialized configuration is advantageous for minimizing uncontrollable interference in congested frequency bands during the experimental process. We fix the reception power at 0 dBm and only change the reception position.

Interference Source: We employ the USRP B210 in conjunction with SDR to simulate a variety of interference sources, encompassing GWN, broadband interference such as WiFi signals, and narrowband signal interference characterized by ZigBee signals. The generation of GWN is facilitated by the Noise Source module of GNU Radio. Meanwhile, the WiFi signals and ZigBee signals are realized based on open-source SDR projects [29], [30].

Setup: We compare our method with Symphony and RSBLE. Symphony is actually the native BLE protocol augmented with symbol extension, whereas RSBLE integrates the native BLE with RS coding. To facilitate a comparable evaluation, we equate the number of BLE bits introduced by symbol expansion in Symphony and RS encoding in RSBLE to the total number of BLE bits introduced by the physical layer and the link layer in our approach. To neutralize the effects stemming from hardware disparities, we implement all three methods with CC2400. We employ an SDR as the receiver, specifically utilizing the USRP B210 to capture the signals from the three methods under comparison. In pursuit of enhancing the instructiveness and reproducibility of our experimental results, the carrier frequency is set to 2.39 GHz to avoid burst interference from the crowded 2.4 GHz band.

Experimental site: The evaluation of the performance of *CPL* over weak links mainly focuses on the physical layer and link layer metrics (e.g., throughput) in weak link communication. We conduct comprehensive experiments in an enclosed conference room at a Line-of-Sight (LOS) site to minimize environmental variables and external interference, thereby providing a consistent backdrop for our experiments. Fig. 8 visually presents the experimental setup within an enclosed conference room, offering a schematic overview of the equipment layout and the LOS conditions maintained throughout our trials. When validating our theoretical model, we initially fix the positions of the transmitter, receiver, and interference source. With the transmitter's power held constant, we vary the power of the interference source to alter the SNR levels. We then compare the throughput measured in the real-world environment at different SNR levels with the throughput calculated by our theoretical model to validate the accuracy of our method. When assessing performance under varying interference conditions, the positions of the transmitter and the interference source are held

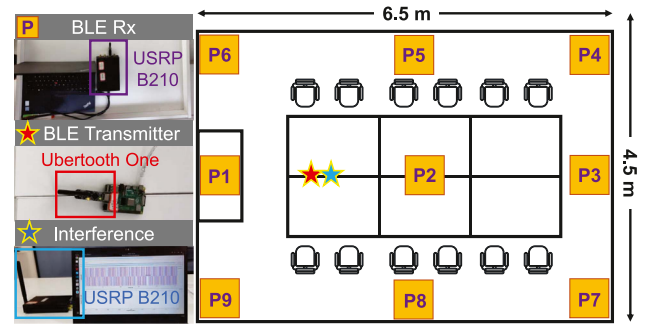


Fig. 8. Experimental testbed and site.

constant, while the receiver is positioned at various locations as depicted in Fig. 8 to measure the throughput.

VII. EVALUATION

A. Validation With Real-World

Prior to the application of the main methods of *CPL*, we conduct experiments to validate our theoretical model by comparing the model output and the results measured in the real-world environment. We set up various schemes with the consistent total number of BLE bits introduced by symbol expansion in the physical layer and coding in the link layer. For instance, these schemes can be differentiated by the expansion of symbols at the physical layer by eight times, four times, two times, and no expansion, corresponding to the link layer not adding extra RS coding, RS coding efficiency of 1/2, 1/4, and 1/8, respectively. Fig. 10 shows the mean absolute error between the actual throughput in real experiments and the theoretical value across various SNR levels. The black lines on the bars in Fig. 10 indicate the standard deviation across multiple trials.

Across all allocation schemes and SNRs, the average absolute error does not exceed 5 (less than 0.98% of the maximum throughput). The standard deviation of the absolute error does not exceed 1.7. Upon further comparison of Fig. 10 with Fig. 6, it is observed that the data points with larger average absolute errors and standard deviations in Fig. 10 generally correspond to the points in Fig. 6 where the throughput increases more sharply with increasing SNR. This is attributed to the higher sensitivity of throughput to SNR changes at these points. Minor errors can have a more significant impact due to the larger slopes. The existence of such situations is reasonable and further substantiates that, within the permissible range of error, the theoretical modeling method used in this paper yields results that closely align with actual outcomes.

B. Optimization of Throughput

To evaluate the optimization of network throughput for *CPL*, we first have the Bluetooth device communicate on a weak link with ambient background noise (GWN as sources of interference). With the positions of the transmitter held constant, we vary the position of the receiver as described in Fig. 8. The

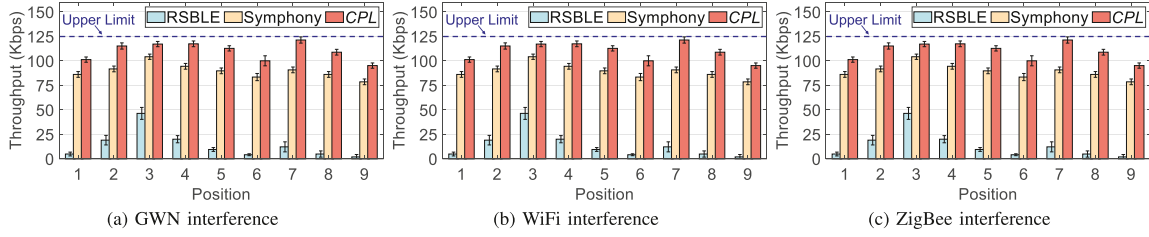


Fig. 9. Comparison of throughput across different methods under different interference.

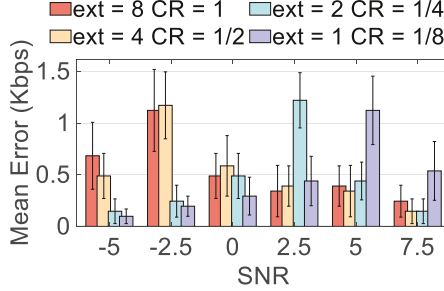


Fig. 10. Experimental method validation with error analysis.

TABLE II
POSITIONAL SNR (dB) UNDER THREE INTERFERENCE TYPES

Type	P1	P2	P3	P4	P5	P6	P7	P8	P9
GWN	1.33	3.08	4.95	2.42	1.35	0.25	3.15	2.01	1.28
WiFi	1.36	3.56	5.21	1.84	0.52	-1.45	3.39	2.22	1.59
ZigBee	0.81	2.06	3.51	2.47	1.92	1.05	3.2	2.1	1.33

throughput measured and calculated at different positions of receiver nodes is depicted in Fig. 8, with the corresponding SNRs presented in Table II.

Fig. 9(a) shows the performance in the presence of GWN interference. when comparing *CPL* to *RSBLE*, the enhancement in throughput ranges from $1.52\times$ to $47.75\times$. Additionally, when comparing *CPL* to *Symphony*, the throughput improvement is noted to be 12.41%–33.60%. Our method and *Symphony* demonstrate a significant performance improvement over *RSBLE*. In comparison with *Symphony*, our method also shows a Considerable improvement in throughput. This improvement is mainly due to the balance between the optimization of the physical layer and the link layer in *CPL*. *CPL* enhances the resilience to interference through a more fine-grained BLE bits allocation strategy. We also calculate their theoretical throughput using (18), as well as the modeling results from the physical layer and link layer (Section IV-A and IV-B). The theoretical average improvements achieved by *CPL* are 22.78% over *Symphony* and $14.44\times$ over *RSBLE*. These theoretical values closely match the actual measured averages of 22.92% over *Symphony* and $15.91\times$ over *RSBLE*, further validating the effectiveness of our approach. We will show more details and analytical support for the system's implementation results in Section VII-D

We also observe that in certain instances, the standard deviation (The black lines on the bars in Fig. 9) of throughput

across multiple trials is notably larger. This phenomenon can be attributed to the sensitivity of throughput to SNR in the prevailing channel conditions, corresponding to the points with a steeper slope in Fig. 6. Even minor fluctuations in the channel environment can lead to significant deviations in the results compared to other points.

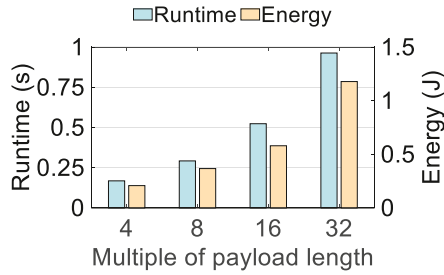
C. Robustness Under Interference

To assess the robustness of the *CPL* under other interference, we conduct experiments under both broadband and narrowband interference conditions. We sequentially employ WiFi signals, and ZigBee signals as sources of interference. With the positions of the transmitter and interference sources held constant, we vary the position of the receiver as described in Fig. 8. The throughput measured and calculated at different interference types and the positions of receiver nodes are depicted in Fig. 8, with the corresponding SNRs presented in Table II.

Fig. 9(b) shows the performance in the presence of WiFi interference. When comparing *CPL* to *RSBLE*, the range of throughput enhancement is $1.25\times$ – $50.56\times$. The throughput improvement of *CPL* relative to *Symphony* ranges from 5.49% to 18.78%. Fig. 9(c) shows the performance in the presence of ZigBee interference. the throughput improvement of *CPL* over *RSBLE* ranges from $5.06\times$ to $27.25\times$. When compared to *Symphony*, the improvement of *CPL* ranges from 25.54% to 32.16%.

CPL can sustain efficiency and robustness under WiFi interference conditions, attributed to its sophisticated redundancy mechanisms and adaptive control strategies. *CPL* exhibits better performance under ZigBee interference compared to that under GWN and WiFi interference. This enhanced performance is attributed to the interference-aware controlled CFO fine-tuning method proposed by *CPL*. It utilizes the controllable CFO of BLE devices to actively mitigate the impact of narrowband interference.

In Fig. 8, P4-P6 are placed symmetrically to P7-P9 with respect to the transmitter and the interference source, serving as the axis of symmetry. However, the throughput measurements at Points P4-P6 are consistently lower than those at Points P7-P9. This discrepancy arises despite the experimental site being set within an enclosed space, which is not electromagnetically isolated from the external environment. In comparison to P7-P9, P4-P6 are in closer proximity to a laboratory equipped with a substantial number of electronic devices, including medium-scale servers, numerous IoT devices, and PCs.

Fig. 11. Offline modeling overhead of *CPL*.

Consequently, the environmental background noise level within the experimental site is not uniformly distributed, with the noise floor at P4-P6 being higher than at P7-P9.

D. In-Depth Analysis of Optimization

Next, we compare the throughput of three methods under different interference types and intensity levels. Fig. 12 shows the results. The interference strength is adjusted by increasing or decreasing the power of the interference source by 60%. We also measure the identification accuracy of different interference types in these complex environments, achieving a high accuracy of 97.22%. Under strong interference scenarios, as shown in Fig. 12(a)–(c), *CPL* generally outperforms Symphony across most test conditions, achieving average throughput improvements of 9.91% under GWN, 8.86% under WiFi interference, and 31.67% under ZigBee interference. However, the RSBLE method demonstrates severely degraded communication performance under strong interference scenarios, failing to maintain normal operation and showing significantly inferior performance compared to *CPL* and Symphony. Position 6 in Fig. 12(a), Position 5 and 6 in Fig. 12(b), *CPL* deteriorates to Symphony. This phenomenon occurs because maximum physical layer symbol expansion using all BLE bits becomes the best strategy under extreme interference conditions. Under low-intensity interference scenarios, as shown in Fig. 12(d)–(f) *CPL* maintains its performance superiority with average throughput enhancements of 9.88%, 8.86%, and 12.04% over Symphony for GWN, WiFi, and ZigBee interference types respectively. Furthermore, *CPL* demonstrates substantial performance gains over RSBLE, achieving throughput improvements of 1.19×, 2.14×, and 1.16× for GWN, WiFi, and ZigBee interference types respectively.

To further investigate the underlying reasons for the significant throughput optimization of our proposed method and identify the key factors limiting throughput under such conditions, we measure the PDR_e and BER_L of the three methods under high-intensity WiFi interference. As illustrated in Fig. 13, these experimental results, when analyzed through the (18), constitute rigorous mathematical validation of our method's superiority. The quantitative framework established by the (18) enables systematic evaluation by incorporating actual measured BER_L and PDR_e values (along with predetermined CR and ext parameters), thereby providing theoretical grounding for performance comparisons. Under strong WiFi interference, RSBLE

TABLE III
ENERGY CONSUMPTION FOR TRANSMITTING 255B DATA (μJ)

Method	P1	P3	P5	P7	P9
CPL	7.97	6.89	7.16	6.66	8.48
Symphony	9.37	7.75	8.99	8.89	10.28
RSBLE	165.45	17.42	84.84	66.18	413.61

significantly degraded PDR_e due to its lack of preamble and AA extension, resulting in substantial packet loss during reception. This severely constrains its communication performance. The Symphony method, which implements the highest extension factor for the preamble and AA, achieves the highest PDR_e . In contrast, *CPL* strategically allocates BLE bits, primarily for physical layer symbol extension while reserving a portion for link layer error correction. Compared to the other two methods, *CPL* maintains a competitive PDR_e while achieving the lowest BER_L , thereby delivering best network throughput. Furthermore, our experimental results reveal that *CPL* performs identically to Symphony at Positions 5 and 6, indicating that *CPL* degenerates to Symphony's strategy in these scenarios. This phenomenon occurs because the interference intensity at these positions is sufficiently severe that conventional BLE packets cannot be reliably received by the receiver. Consequently, PDR_e becomes the dominant factor limiting communication performance. Under such extreme conditions, the most effective strategy is to adopt Symphony's approach, which maximizes the physical layer extension factor to ensure reliable packet reception. This dual validation approach, which combines empirical measurements with analytical modeling, conclusively demonstrates that our cross-layer optimization strategy achieves best throughput through intelligent BLE bit allocation, which simultaneously maintains competitive PDR_e levels and minimizes BER_L .

E. Overhead of System

Transmitter consumption: We evaluate the energy consumption of three methods by recording their radio duration and calculating the energy consumption using voltage and current values from the Ubereooth One user documentation [31]. Table III shows the average energy consumption required to reliably deliver 255 Bytes of data. *CPL* reduces energy consumption by 11.04% to 25.15% compared to Symphony, and 60.42% to 97.95% compared to RSBLE.

Offline modeling overhead: We evaluate the computational overhead and energy consumption of offline modeling. Fig. 11 shows the runtime and energy consumption of offline modeling for BLE packets with various multiples of payload lengths after physical layer expansion and link layer encoding. The offline modeling program is executed on a Raspberry Pi 4 Model B, with the execution time measured using the time module from the Python standard library. We employ a USB power meter [32] to measure the additional power consumption incurred by running the offline modulation program. The runtime does not exceed 1 s, and the energy consumption is capped at 1.2 Joules. With low overhead, the system is capable of operating

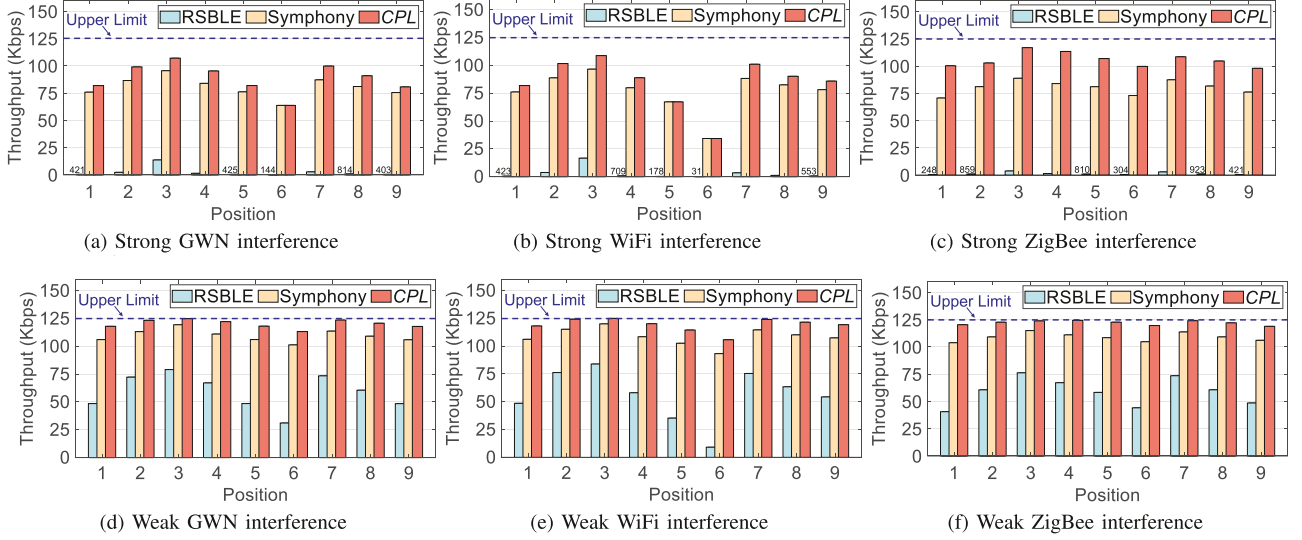


Fig. 12. Comparison of the throughput for three methods under varying types and levels of interference. The level of interference is altered by increasing or decreasing the power of the interference source by 60%. In the bar graph, due to the small values being difficult to discern, the units for these values are noted as bits per second (bps).

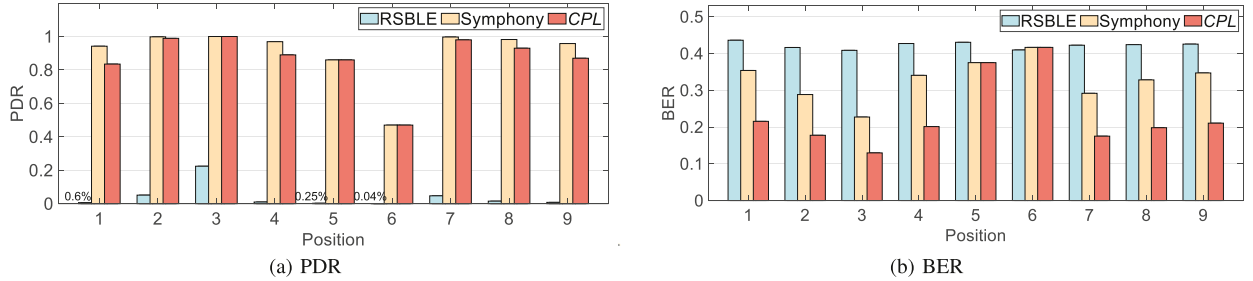


Fig. 13. PDR and BER under strong WiFi interference.

on resource-constrained devices. As the multiples of payload length increase, the runtime and energy consumption of offline modeling escalate. To facilitate the implementation of physical layer symbol extension on the Ubetooth One, we constrain the expansion multiples to integer values. Consequently, as the multiples of payload length grow larger, a greater number of schemes become available for BLE bits allocation by the physical layer and the link layer. Offline modeling necessitates the computation of throughput for each scheme under varying SNRs. Hence, an increased total number of schemes demands more runtime and energy consumption.

Online prediction overhead: We subsequently evaluate the time overhead of CPL's online prediction and compare it with existing methods such as 4C [8] and TALENT [9]. The online prediction program is executed on the receiver, and the execution time is measured using the time module from the Python standard library. Fig. 14 illustrates the time overhead required for the three methods to predict the effective data transmission rate under different allocation schemes and determine the optimization scheme for nodes 1, 3, 5, 7, and 9. Compared to 4C, CPL achieves a reduction in time overhead ranging from

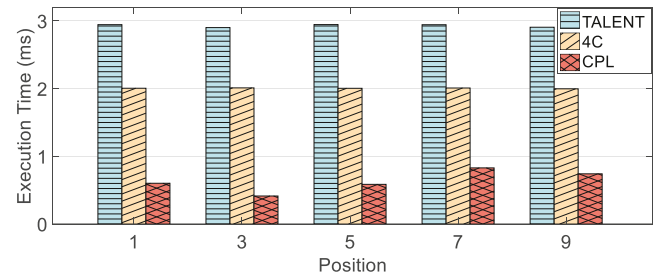


Fig. 14. Online prediction overhead.

58.71% to 79.40%, and compared to TALENT, the reduction is from 71.82% to 85.72%. On the one hand, 4C employs machine learning to construct its model, necessitating time-consuming vector multiplication and sigmoid function calculations for predicting the effective data transmission rate of the next packet. Even with the use of a pairwise linear approximation [33] to approximate the sigmoid function, 4C still requires approximately 0.5 milliseconds for each prediction. On the other hand, TALENT uses a stochastic gradient descent (SGD) [34] online

learning algorithm to train its model without the need for training data collection as in 4C, but this introduces additional computational overhead for periodic weight updates. Our method, *CPL*, does not require complex and time-consuming calculations during online prediction nor periodic weight updates, significantly reducing the time overhead in the online prediction phase.

Allocation adjusting overhead: The overhead introduced by allocation adjusting mainly consists of the time spent on the receiver's periodic channel monitoring, the transmission and reception of BLE packets, and the mode switching on the Ubertooth device. There exists a trade-off between the sensing frequency and its precision. Based on observations from IoT deployments in daily scenarios, we set the sensing interval to five seconds, as the environment in typical household IoT applications [35], [36], primarily affected by multipath effects, generally does not undergo significant changes within this time frame. Our SNR-based optimization model shows minimal performance impact from normal SNR fluctuations, with sufficiently spaced switching thresholds (>3 dB) to prevent unnecessary switching of the scheme. Although the reduced sensing frequency marginally sacrifices update responsiveness, this design choice effectively balances overhead reduction with acceptable accuracy in practical deployments. Supplementary experiments have revealed that the receiver performs channel monitoring every 5 seconds, involving the inspection of 1024 samples, with an average time cost of 85.2 microseconds. The computational cost of channel monitoring in the provided code is primarily due to the complex 1024-point operation, which dominates with an algorithmic complexity of $O(N \log N) = 10,240$ ($N=1024$). Auxiliary steps such as frequency axis generation and noise power summation contribute to the $O(N)$ complexity. Memory usage peaks at approximately 20 KB per iteration due to the storage of complex-valued input signals (8 KB), FFT results (8 KB), and frequency arrays (4 KB). Although resource-constrained devices like the nRF52840 (with 1 MB flash) must manage memory with care, the 20 KB overhead per FFT iteration is negligible, accounting for only 1.95% of total flash. The mode switching time for Ubertooth nodes ranges from 1.14 to 10.29 milliseconds, and the transmission time for BLE packets between the BLE transmitter and receiver is less than $160 \mu s$. Despite these additional overheads, the overall impact on system efficiency remains minimal. These extra steps ensure that the system can dynamically adapt to changes in the allocation scheme without significantly degrading performance. This method of managing changes in the scheme is an integral part of the system design, striking a balance between the need for flexibility and the maintenance of efficient communication.

F. Scalability in Multi-Device Environments

The prior studies [37], [38], [39] show native BLE failed to provide reliable service for those multiple connection applications due to packet loss and collisions. There are some existing works such as BLEX [40] already address this challenge. BLEX is a link layer connection manager implemented on top of the Zephyr [41] controller in multi-device environments. This manager dynamically adjusts the anchor points of connection events

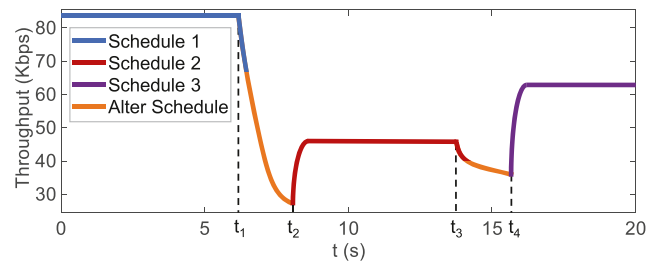


Fig. 15. Dynamic optimization by *CPL* in varying channels.

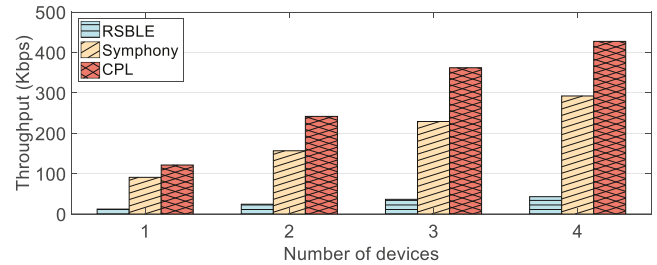


Fig. 16. Scalability of the system in multi-device environments.

to efficiently mitigate interference of multiple concurrent devices. *CPL* allows modifications to link layer configurations that exhibit superior compatibility with BLEX. *CPL* is orthogonal and non-conflicting with these works so *CPL* can be seamlessly extended to multi-device scenarios to further optimize weak link performance.

As RSBLE also allows modification of link layer configurations, we incorporate both RSBLE and *CPL* with BLEX to evaluate their performance in multi-device environments. In contrast, Symphony modifies physical-layer parameterizations while restricting access to link layer reconfigurations, thus failing to incorporate with BLEX. We only apply symphony with Zephyr Os in following experiments for comparison. To compare the scalability in multi-device environments between the three methods, we conduct the following experiments with an increasing number of BLE devices (e.g., Nordic nrf52840 [11] in the same traffic scenario). Fig. 16 shows the measured throughput. With the link layer connection manager, *CPL* and RSBLE schemes demonstrate less than 1% throughput degradation even when one or two newly joined devices with identical configurations join the BLE multi-device network. The connection manager dynamically adjusts anchor points based on traffic load and idle time to prevent collisions and interference. For increased traffic, it moves anchor points forward to allocate additional resources; for excessive idle time, it delays anchor points to consolidate idle resources. However, when three devices join the network, both schemes experience a throughput degradation of approximately 12%. The reason for this is that the connection manager fails to meet real-time latency requirements, as it can only adjust a subset of connection parameters. It also struggles with rapid connection re-scheduling, especially in updating the anchor point position, connection interval, and event length. A mandatory delay of six connection intervals further degrades transmission performance

in dynamic traffic conditions. Due to its inherent limitations, Symphony is unable to utilize the link layer connection manager, resulting in a significant reduction in throughput when new devices join the network. Specifically, the throughput degradation is observed to be 13.8%, 15.9%, and 19.6% respectively when one, two, and three new devices are connected. This is because Zephyr OS and Nordic employ simplistic and independent connection scheduling mechanisms that fail to consider the status and requirements of previously connected peripherals when establishing new connections. These systems prioritize rapid connection establishment through preemptive scheduling based on priority policies, often leading to resource overlapping issues where newly connected devices encroach upon resources allocated to existing connections.

G. Case Study

We further illustrate how *CPL* dynamically adapts to channel changes through a case study. We measure the throughput of the BLE nodes at different time periods. We set the initial channel SNR to -0.5 dB and the noise pattern to GWN. At t_1 , we increase the noise power to set the SNR to -3 dB. At t_2 , we modify the noise pattern to a narrowband signal (ZigBee) of the same power.

Fig. 15 shows the variation of throughput of the BLE nodes in the case under different time periods. Initially, *CPL* is engaged in an best configuration tailored to the prevailing channel conditions under broadband interference, represented by the blue solid line. At t_1 , the channel environment undergoes a sudden degradation, precipitating a sharp decline in throughput. *CPL*, equipped with real-time channel monitoring capabilities, promptly initiates a transition to a configuration better suited for the adverse channel conditions, as indicated by the orange solid line. *CPL* completes the conversion at t_2 , within a duration not exceeding 2 seconds. Subsequently, at t_3 , the interference shifts to a narrowband signal, leading to a further reduction in throughput. Through spectral analysis, *CPL* detects the presence of the narrowband interference and employs an interference-aware controlled CFO fine-tuning technology that leverages controllable CFO to mitigate its effects. Specifically, *CPL* adjusts the registers within the Ubertooth One device that controls the CFO, modulating the CFO of the BLE signal to effectively circumvent the narrowband signal and mitigate its interference. At the t_4 , *CPL* completes the conversion within a duration not exceeding 2 seconds.

The case study demonstrates that *CPL* dynamically adapts to channel changes by promptly adjusting its configuration in response to varying SNR levels and noise patterns. *CPL* can effectively mitigate interference and optimize throughput within a 2-second timeframe.

VIII. CONCLUSION

In this paper, we propose *CPL*, a novel Cross-Layer Optimization technique, which significantly boosts the performance of BLE, particularly in weak link scenarios. Our approach dynamically combines physical layer symbol extension with link layer coding, demonstrating a substantial throughput enhancement

under various interference conditions. We implement *CPL* with COTS BLE chips and SDR. Extensive experiments confirm its superior throughput and robustness in real-world applications, highlighting its potential for IoT deployments. Compared with existing work, *CPL* demonstrates a competitive advantage by achieving 5.49%–18.78% higher throughput under identical configurations. With interference-aware controlled CFO fine-tuning technique, *CPL* can adapt to narrowband interference and has 25.54%–32.16% higher throughput.

REFERENCES

- [1] B. SIG, "Bluetooth market update," Website, 2024, [Online]. Available: <https://www.bluetooth.com/2024-market-update/>
- [2] M. Spörk, C. A. Boano, and K. Römer, "Improving the timeliness of bluetooth low energy in noisy RF environments," in *Proc. ACM Int. Conf. Embedded Wireless Syst. Netw.*, 2019, pp. 23–34.
- [3] M. Spörk, C. A. Boano, and K. Römer, "Improving the timeliness of bluetooth low energy in dynamic RF environments," *ACM Trans. Internet Thing*, vol. 1, no. 2, pp. 1–32, 2020.
- [4] Z. Li and Y. Chen, "Achieving universal low-power wide-area networks on existing wireless devices," in *Proc. IEEE 27th Int. Conf. Netw. Protoc.*, 2019, pp. 1–11.
- [5] C. Li et al., "ChirpTransformer: Versatile LoRa encoding for low-power wide-area IoT," in *Proc. 22nd Annu. Int. Conf. Mobile Syst. Appl. Serv.*, 2024, pp. 479–491.
- [6] C. Şişman, S. Sağır, İ. Kaya, S. Ünal, and Y. Baltacı, "Coding and diversity gains of low power radio communications: BLE and ZigBee," in *Proc. IEEE 41st Int. Conf. Telecommun. Signal Process.*, 2018, pp. 1–4.
- [7] Z. Li, W. Du, Y. Zheng, M. Li, and D. Wu, "From rateless to hopless," in *Proc. ACM Des. Mobile Netw. Mobile Comput.*, 2015, pp. 107–116.
- [8] T. Liu and A. E. Cerpa, "Foresee (4C): Wireless link prediction using link features," in *Proc. 10th ACM/IEEE Int. Conf. Inf. Process. Sensor Netw.*, 2011, pp. 294–305.
- [9] T. Liu and A. E. Cerpa, "Temporal adaptive link quality prediction with online learning," *ACM Trans. Sensor Netw.*, vol. 10, no. 3, pp. 1–41, 2014.
- [10] W. Dong et al., "DPLC: Dynamic packet length control in wireless sensor networks," in *Proc. IEEE Int. Conf. Comput. Commun.*, 2010, pp. 1–9.
- [11] NORDIC, "NRF52840 product specification v1.8," Website, 2024. [Online]. Available: https://infocenter.nordicsemi.com/pdf/nRF52840_PS_v1.8.pdf
- [12] T. Instruments, "2.4-GHz bluetooth low energy system-on-chip," Website, 2024. [Online]. Available: <https://www.ti.com.cn/cn/lit/ds/symlink/cc2540.pdf>
- [13] Y. Cao, L. Kong, L. He, G. Chen, M.-Y. Wu, and T. He, "Mrs. Z: Improving zigbee throughput via multi-rate transmission," in *Proc. IEEE 25th Int. Conf. Netw. Protoc.*, 2017, pp. 1–10.
- [14] J. S. Roh, "Performance analysis and evaluation of bluetooth networks in wireless channel environment," in *Proc. IEEE Int. Conf. Syst. Netw. Commun.*, 2006, pp. 61–61.
- [15] I. S. Reed and G. Solomon, "Polynomial codes over certain finite fields," *J. Soc. Ind. Appl. Math.*, vol. 8, no. 2, pp. 300–304, 1960.
- [16] R. G. Gallager, *Information Theory and Reliable Communication*, vol. 588. Berlin, Germany: Springer, 1968.
- [17] S. Bernard, *Digital Communications: Fundamentals and Applications*. London, U.K.: Pearson, 2021.
- [18] The Bluetooth Special Interest Group, *Specification of the Bluetooth System: Core System Package [Low Energy Specification], Part A: Physical Layer Specification*, 5th ed. Kirkland, WA, USA: The Bluetooth Special Interest Group, 2016, [Online]. Available: <https://www.bluetooth.org/>
- [19] W. Kester, "MT-001: Taking the mystery out of the infamous formula," *snr* = $6.02 n 1.76$ db," and why you should care," *REV. 0*, 2005, 10–03.
- [20] W. Yan, T. Voigt, and C. Rohner, "RRF: A robust radiometric fingerprint system that embraces wireless channel diversity," in *Proc. 15th ACM Conf. Secur. Privacy Wireless Mobile Netw.*, 2022, pp. 85–97.
- [21] J. Cheon, H. Hwang, D. Kim, and Y. Jung, "IEEE 802.15.4 ZigBee-based time-of-arrival estimation for wireless sensor networks," *Sensors*, vol. 16, no. 2, 2016, Art. no. 203.
- [22] J. Hua, H. Sun, Z. Shen, Z. Qian, and S. Zhong, "Accurate and efficient wireless device fingerprinting using channel state information," in *Proc. IEEE Int. Conf. Comput. Commun.*, 2018, pp. 1700–1708.

- [23] M. Ryan, "Ubertooth," Website, 2024. [Online]. Available: <https://github.com/greatscottgadgets/ubertooth/tree/master>
- [24] LPC176X/5X user manual, Website, 2024. [Online]. Available: https://www.nxp.com/docs/en/data-sheet/LPC1759_58_56_54_52_51.pdf
- [25] T. Instruments, "CC2400 2.4 GHz low-power RF transceiver," Website, 2024. [Online]. Available: <https://www.ti.com/lit/ds/symlink/cc2400.pdf?ts=1708517236243>
- [26] T. Instruments, "CC2591 2.4-GHz RF front end," Website, 2024. [Online]. Available: <https://www.ti.com.cn/cn/lit/ds/symlink/cc2591.pdf>
- [27] H.-W. Cho and K. G. Shin, "FLEW: Fully emulated WiFi," in *Proc. ACM Annu. Int. Conf. Mobile Comput. Netw.*, 2022, pp. 29–41.
- [28] B. SIG, "Core specification 5. 4," Website, 2024. [Online]. Available: <https://www.bluetooth.com/specifications/specs/core-specification-5-4/>
- [29] P. U. Bastian Bloessl, "GR-IEEE802-11," Website, 2024. [Online]. Available: <https://github.com/bastibl/gr-ieee802-11>
- [30] B. Bloessl, "GR-IEEE802-15-4," Website, 2024. [Online]. Available: <https://github.com/bastibl/gr-ieee802-15-4>
- [31] Ubertooth User Documentation, Website, 2024. [Online]. Available: https://ubertooth.readthedocs.io/_/downloads/en/latest/pdf/
- [32] ChargerLAB, "KM003C/KM002C USB tester user manual," Website, 2024. [Online]. Available: https://docs.google.com/document/d/1T55UeXUNN9La3tzAF0jO45Snw_rdr9RVGt3XQeRIM8
- [33] H. Amin, K. M. Curtis, and B. R. Hayes-Gill, "Piecewise linear approximation applied to nonlinear function of a neural network," *IEEE Proc. -Circuits, Devices Syst.*, vol. 144, no. 6, pp. 313–317, 1997.
- [34] T. M. Mitchell, *Machine Learning*, vol. 1. New York, NY, USA: McGraw-Hill, Inc., 1997.
- [35] J. Yang, C. Poellabauer, P. Mitra, and C. Neubecker, "Beyond beaconing: Emerging applications and challenges of BLE," *Ad Hoc Netw.*, vol. 97, 2020, Art. no. 102015.
- [36] K. E. Jeon, J. She, P. Soonsawad, and P. C. Ng, "BLE beacons for Internet of Things applications: Survey, challenges, and opportunities," *IEEE Internet Things J.*, vol. 5, no. 2, pp. 811–828, Apr. 2018.
- [37] M. Kim and J. Paek, "Multi-connection scheduling for resource fairness in bluetooth low energy networks," in *Proc. 14th Int. Conf. Inf. Commun. Technol. Convergence*, 2023, pp. 533–535.
- [38] M. Kim and J. Paek, "Multi-connection scheduling based on connection subrating for fair resource allocation in bluetooth low energy networks," in *Proc. 21st ACM Conf. Embedded Networked Sensor Syst.*, 2023, pp. 550–551.
- [39] Y. Li, B. Li, J. Lv, and W. Dong, "BLEdge: Edge-centric programming for BLE applications with multi-connection optimization," *ACM Trans. Sensor Netw.*, vol. 20, no. 6, pp. 1–25, 2024.
- [40] E. Park, H.-S. Kim, and S. Bahk, "BLEX: Flexible multi-connection scheduling for bluetooth low energy," in *Proc. ACM/IEEE 20th Int. Conf. Inf. Process. Sensor Netw.*, 2021, pp. 268–282.
- [41] A. Aillet, "Zephyr project," 2024. [Online]. Available: <https://zephyrproject.org/>



Renjie Li received the BS degree from Dongbei University, China, in 2024. He is currently working towards the PhD degree with the College of Computer Science, Zhejiang University, China. His research interests include wireless networking and edge computing.



Yeming Li received the BS degree in computer science from the Zhejiang University of Technology, Hangzhou, in 2020. He is currently working toward the PhD degree with the College of Computer Science, Zhejiang University, China. His research interests include the Internet of Things and wireless communication protocols.



Jiamei Lv received the BS degree from the College of Information Science and Engineering, Ningbo University in 2018 and the PhD degree from the College of Computer Science, Zhejiang University in 2023. She is currently a researcher with the School of Software Technology, Zhejiang University. Her research interests include Internet of Things, edge computing, and 5G networks.



Hailong Lin received the BE degree from Xi'an Jiaotong University, China, in 2022. He is currently working towards the master's degree with the College of Computer Science, Zhejiang University, China. His research interests include wireless networking and IoT security.



Yi Gao (Member, IEEE) received the BS and PhD degrees from the College of Computer Science, Zhejiang University, China, in 2009 and 2014, respectively. From 2015 to 2016, he visited McGill University as a visiting scholar. He is currently a full professor with the College of Computer Science, Zhejiang University. His current research interests include Internet of Things and intelligent edge computing. He is a member of the ACM.



Wei Dong (Member, IEEE) received the BS and PhD degrees from Zhejiang University in 2005 and 2011, respectively. He is currently a full professor with the College of Computer Science, Zhejiang University. He leads the Embedded and Networked Systems (Em-Nets) lab in Zhejiang University. He has published more than 100 papers in prestigious conferences and journals including MobiCom, INFOCOM, ICNP and *IEEE/ACM Transactions on Networking*, *IEEE Transactions on Mobile Computing*, etc. His research interests include Internet of Things and sensor networks, wireless and mobile computing, and network measurement. He is a member of ACM.

works, wireless and mobile computing, and network measurement. He is a member of ACM.